

“Technical Support to Implement Citizen-Led Data Generation and Management Activities”

-Final draft of the inception report-

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List of Acronyms

Acronym	Full Meaning
API	Application Programming Interface
BMUKN	Federal Ministry for the Environment, Nature Conservation and Nuclear Safety
CBOs	Community-Based Organisations
CIP	Conservation Investment Plan
CPUE	Catch Per Unit Effort
CSOs	Civil Society Organisations
GSM	Global System for Mobile Communications
IBA	Important Bird Area
IHE	IHE Delft Institute for Water Education
INGO	International Non-Governmental Organisation
ISO	International Organization for Standardization
KEWASNET	Kenya Water and Sanitation Civil Society Network
MVP	Minimum Viable Product
NBD	Nile Basin Discourse
NBI	Nile Basin Initiative
NDF	National Discourse Forum
NELSAP	Nile Equatorial Lakes Subsidiary Action Programme
NGO	Non-Governmental Organisation
QA/QC	Quality Assurance / Quality Control
SDG	Sustainable Development Goal

STP	Sewage Treatment Plant
ToT	Training of Trainers
UoN	University of Nairobi
USSD	Unstructured Supplementary Service Data
UWASNET	Uganda Water and Sanitation NGO Network
WWF	World Wide Fund for Nature

1.0 Background

Data on wetlands across the Nile Basin remain fragmented among agencies and projects, often outdated or inaccessible to those who need them most. Yet citizens living in and around these wetlands continuously observe ecological change.

Nile Basin Discourse (NBD) has a membership of over 600 CSOs and NGOs organised under ten National Discourse Forums (NDFs) one in each Nile Basin country. NBD provides a unique mechanism for stakeholder engagement at both national and regional levels and uses citizen dialogues to create data ecosystems that directly support planning, management, and policy dialogues, a system where citizen-generated data become credible, structured, and routinely used in wetlands and basin decision-making.

The “Technical Support to Implement Citizen-Led Data Generation and Management Activities” assignment therefore aims to establish the foundation of a data ecosystem through two demonstration pilots Mara Wetlands (Kenya and Tanzania) and Sio-Siteko Wetlands (Uganda and Kenya) representing contrasting ecological and governance settings of the Nile Basin. Each pilot will test how citizen science, academic validation, and institutional anchoring can converge into a credible and replicable model.

Akvo Foundation will lead the execution of the assignment. It will be supported by the following sub consultants in two different wetland sites.

Makerere University (Uganda) and the University of Nairobi (Kenya) will serve as academic partners-co-developing and validating citizen-led monitoring methodologies, ensuring data quality through QA/QC processes, and embedding the approach within academic and institutional systems for long-term sustainability. UWASNET (Uganda Water and Sanitation NGO Network) and KEWASNET (Kenya Water and Sanitation Civil Society Network) will act as civil society anchors, leading community mobilisation, facilitating stakeholder engagement, and ensuring that citizen-generated data is inclusive, locally owned, and effectively linked to governance and policy processes at national and sub-national levels. Together, the consortium combines technical expertise, scientific credibility, and strong community engagement to deliver a robust and scalable citizen-led data ecosystem across both sites.

1.1 About the inception report

A zero draft of the inception report was drafted in January 2026. This report was based on the collective high-level understanding of Akvo, UoN, KEWASNET, UWASNET, and Makerere University. Conversations were also held with end users and water user association representatives. But neither our¹ sample of respondents nor our observations of the on-ground situation were fair representations of the field realities. Secondary data and literature were used to fill the initial information gaps. But secondary literature is not a substitute for experiencing and collecting primary sources of information. The inception report process was therefore temporarily paused. A need was identified to visit the actual locations, trace river pathways, and engage local communities and government representatives before finalizing the report. The current version of the inception report

¹The word ‘our’ or ‘we’ is used in this report for better readability. Unless otherwise stated, it reflects a collective sense of the consortium (Akvo, civil society and academic partners)

is based on the on-ground realities that were observed and documented during field visits. Hence, some parts of the report are written in past tense to share our reflections with the readers. In addition, the revised report dives deep into the methodology, breaks down the data flow and technology architecture, presents the timelines and highlights the risks that may delay/derail our execution with ways to mitigate those risks, and adds the documentation/forms/inputs necessary to execute the assignment.

2.0 Methodology

The methodology will include desk research, reflections on the stakeholder identification and consultation report shared by NBD, reconnaissance visit, stakeholder engagement and debriefing following the field visit, selection of sampling sites, selection of water quality parameters, selection of citizen scientists, deconstructing data workflows and technical architecture for data collection, curation and dissemination. At the time of finalising the inception report, the reconnaissance visit, preliminary site selection, stakeholder engagement, and debriefing had already been completed.

2.1 Desk research

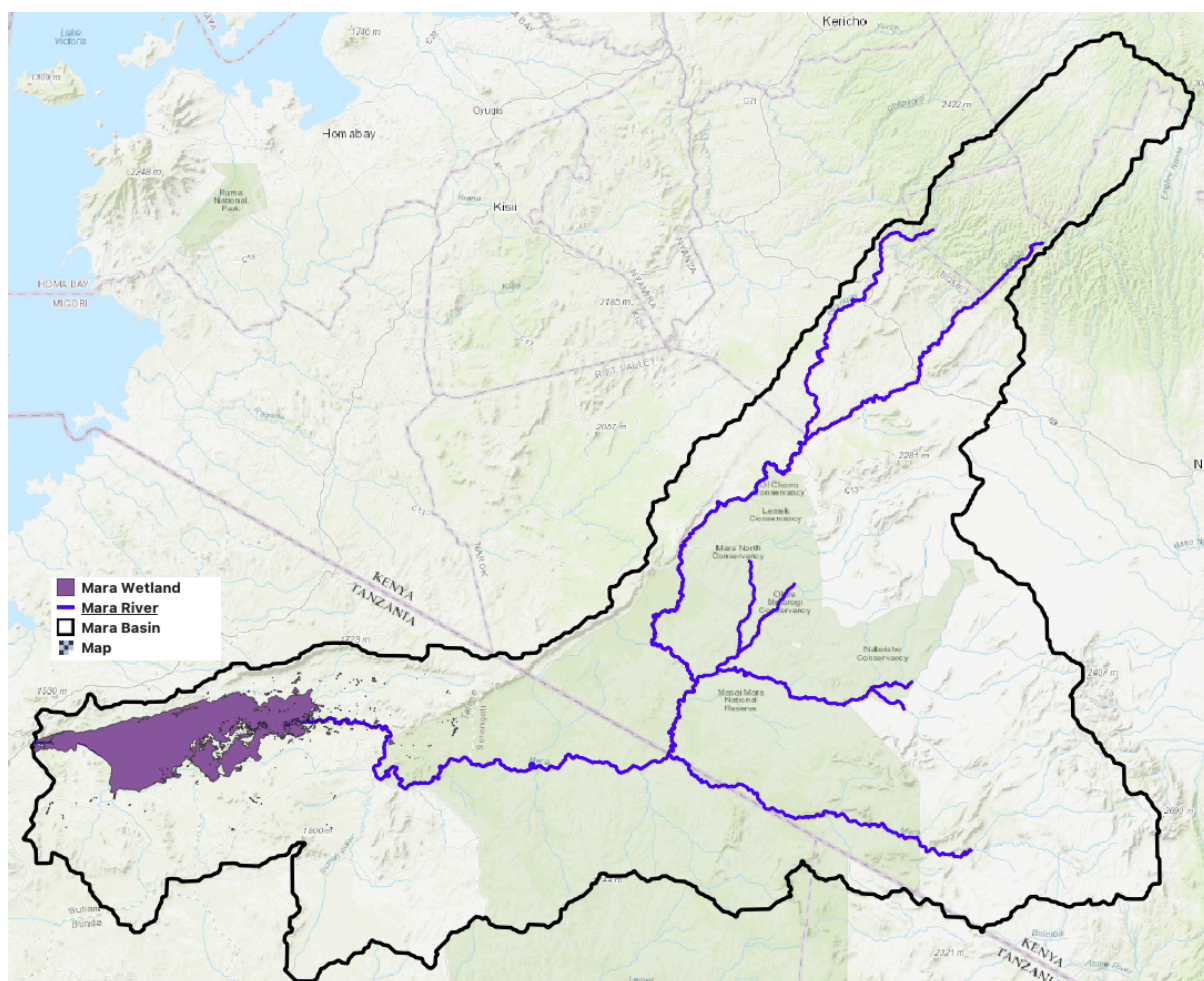
Mara wetlands

The Mara Wetlands form a large papyrus-dominated floodplain of roughly 400–500 km² whose extent varies seasonally. The wetland spans Butiama, Rorya, Tarime, and Serengeti districts and supports an estimated 56,000 people who rely primarily on rain-fed agriculture and small-scale fishing.

This wetland is part of the wider transboundary Mara River Basin, which originates in Kenya's Mau Escarpment and covers approximately 13,325 km², with about 65% of the basin located in Kenya and 35% in Tanzania. The river flows through key ecosystems including agricultural landscapes, the Maasai Mara National Reserve, and the Serengeti before draining into Lake Victoria through the Mara Wetlands. Rapid population growth and land conversion for crops and grazing have transformed more than 100 km² of wetland over the past decade.

Degradation originates upstream in Kenya's Mau Highlands, where deforestation and intensive cultivation drive sedimentation, nutrient loading, and pollution from small-scale mining. Combined with unregulated water abstraction, these pressures have reduced water quality and altered the wetland's natural flow regime. Climate projections indicate rising temperatures ($\approx 1.8^\circ\text{C}$) and higher rainfall (10–12 percent) by 2050, amplifying both flood and drought risks for communities already dependent on unimproved water sources.²

²United States Agency for International Development (USAID), *Vulnerability and Adaptation in the Mara River Basin* (Washington DC: USAID ATLAS Project, 2019), pp. 69–70.



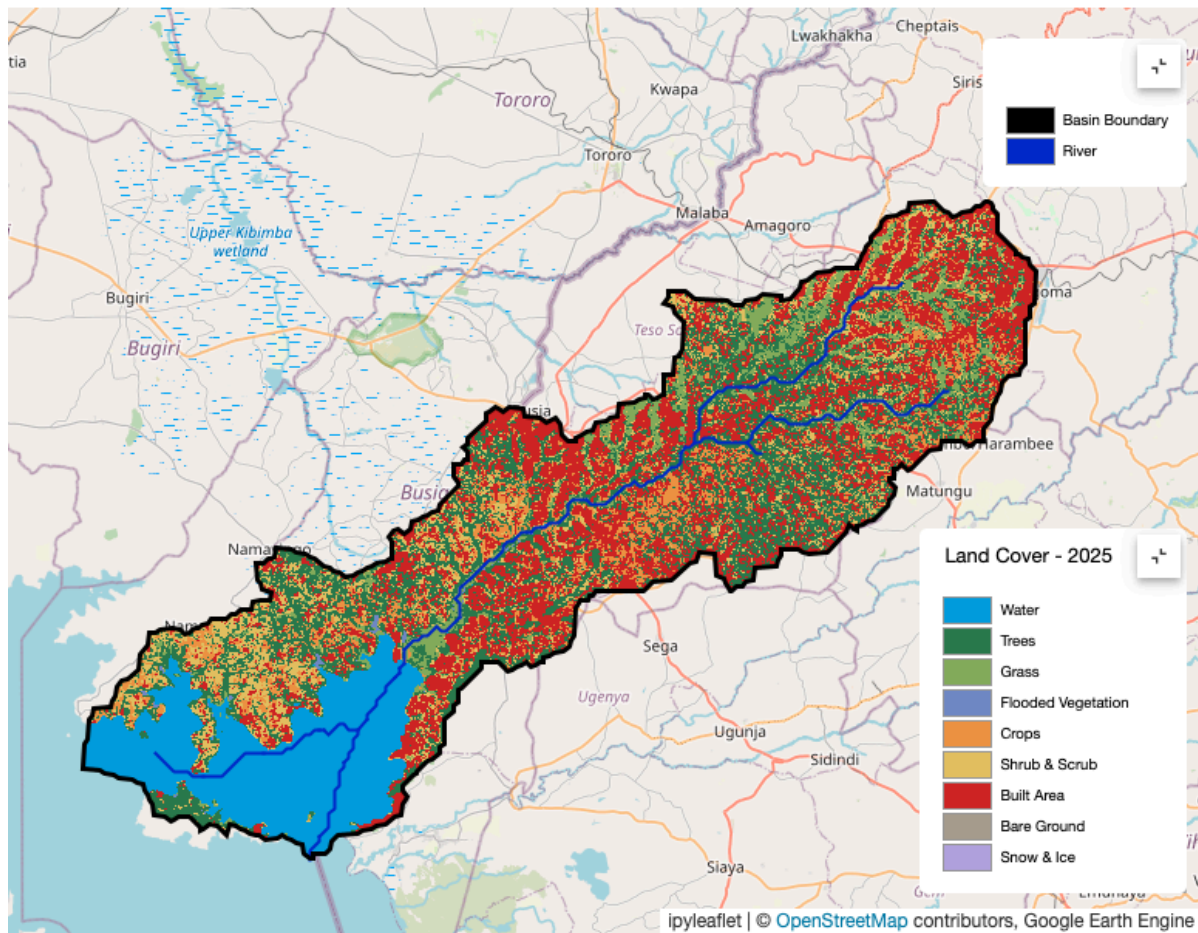
Sio-Siteko wetlands

The Sio–Siteko Wetland Landscape lies north East of Lake Victoria, straddling the Kenya and Uganda border across Busia County (Kenya) and Busia District (Uganda). Covering roughly 415 km² within the lower Sio River catchment (1,437 km²), it regulates river flows, buffers floods, and sustains farming, fishing, grazing, and papyrus-based livelihoods. The wetland is an Important Bird Area (IBA) hosting over 520 bird species alongside diverse fish, plant, and amphibian populations.

Despite its ecological and socio-economic value, the wetland faces rapid degradation from unsustainable land use, vegetation clearing for charcoal, encroachment, and sand harvesting. Overfishing, invasive species such as *Mimosa pudica*, *Lantana camara*, and *Pontederia crassipes*, and weak transboundary coordination have further eroded ecosystem integrity.

The Sio–Siteko Conservation Investment Plan (CIP) (NELSAP–NBI, 2020) and its companion Transboundary Wetland Management Plan (2020–2030) call for stronger institutional coordination, participatory monitoring, and digital information systems. These frameworks provide a sound institutional and policy foundation for piloting citizen-led data collection and ecosystem monitoring under this assignment³.

³Nile Basin Initiative/Nile Equatorial Lakes Subsidiary Action Programme (NELSAP), Sio-Siteko Conservation Investment Plan (CIP) (Entebbe: NBI/NELSAP and Deutsche Gesellschaft für Internationale Zusammenarbeit



2.2. Reflections on the stakeholder identification and consultation reports submitted by NBD

Nile Basin Discourse (NBD) had shared its reports of on-site stakeholder identification and consultation early in the assignment. Based on those reports and subsequent conversations with the NBD team, it became clear that it is better to combine our reconnaissance visit with stakeholder engagement and debriefing on site. However, unlike the previous NBD visit, the process began with field visits during which sampling sites were identified, observations were documented, and a few stakeholders were engaged informally, followed by formal community and technical debriefing sessions.

At the community level, initial stakeholder consultations and focus group discussions were held at each site, with approximately 25 participants per session, including representatives from water user associations, local leaders, and community members. For a technical audience comprising of government, NGOs and academic partners, a similar number of participants was engaged. The details from these consultations and field visits will be available in the field reports written separately for Mara and Sio Siteko wetlands.

Beyond the debriefing during the field visit, it was realized that country level data and information sharing workshops may be necessary at a later stage (once we have adequate citizen-generated data

(GIZ) GmbH, on behalf of the German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety (BMUKN) under its International Climate Initiative (IKI), 2020), pp. 3–5.

available for sharing), with a captive audience including citizens, government, NGOs and academic partners.

The Mara Wetland is located in Tanzania within the wider Mara River Basin, a transboundary system shared between Kenya and Tanzania. As a result, activities upstream in Kenya directly influence downstream river water quality in Tanzania. Similarly, the Sio–Siteko Wetland is also a transboundary ecosystem spanning Kenya and Uganda, with ecological and management dynamics that extend across national borders. Considering both these contexts, transboundary engagements are anticipated, bringing together stakeholders from basin countries (Kenya, Tanzania, Uganda) to support alignment of wetland management actions and share learnings. Whether we will have country level sharing sessions or combine them into a transboundary sharing session, those details will be worked out together with NBD and the key government stakeholders.

Last but not least, during the review of these reports, opportunities for strategic partnerships were actively explored to strengthen the vision of a citizen-led data ecosystem across the pilot locations. Existing citizen-generated data initiatives, particularly on physico-chemical parameters, were identified in the Mara Wetlands, including activities previously supported by Earthwatch. On the ground, these efforts are currently being coordinated by the Lake Victoria Basin Water Board – Mara Catchment, which oversees data collection from enumerators and supports data validation processes.

Similarly, the World Wide Fund for Nature (WWF), in partnership with Tanzania’s Fisheries Research Institute (TAFIRI) and IHE Delft, is undertaking ecological surveys within the Mara Wetlands. As part of this work, a mobile application has been developed to document fish catch per unit effort at selected locations across the wetland.

Local government partners were also engaged during the Tanzania field visit and expressed interest in exploring how this assignment could build on, complement, and learn from the ongoing Earthwatch and WWF initiatives. The intention is to minimise duplication of effort, enhance coordination among stakeholders, and improve overall efficiency.

At this stage, formal engagement with the international partners has not yet commenced. The extent to which the respective programmes may complement one another will therefore depend on the outcomes of these forthcoming discussions.

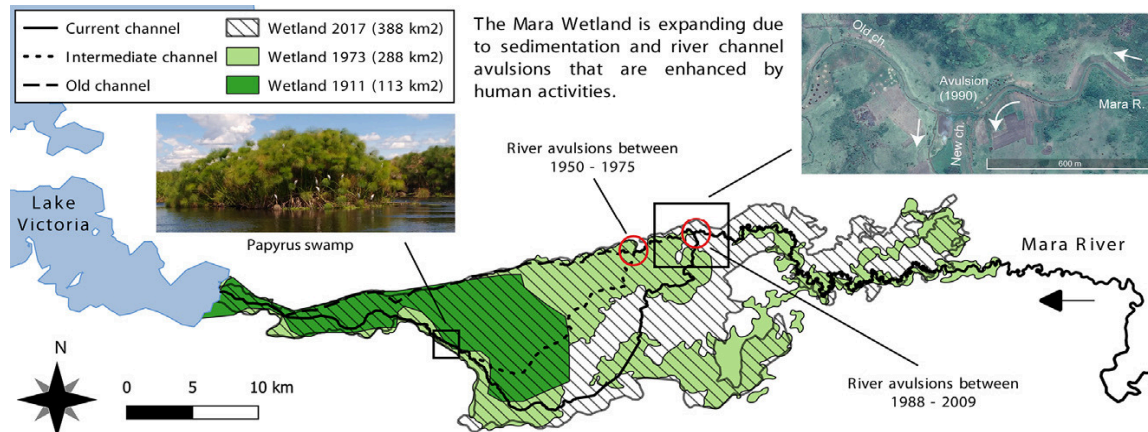
2.3. Reconnaissance visit

Travels were made to the pilot locations to understand the physical and social landscape of the Mara and Sio-Siteko wetlands. Prior to the field visits, available secondary data and NBD stakeholder engagement reports were reviewed. Consultations were also held with existing networks around the wetland sites and developed reconnaissance tools (e.g., site checklists, focus group discussion guides, and mapping templates) in consultation with NBD to guide data collection and ensure consistency during the field assessment.

Discussions were conducted to review and update the stakeholder list previously identified by NBD. Stakeholders were engaged through existing structures and organisations, including Water User Associations and conservation groups (such as community-based organisations and NGOs), while ensuring gender balance to the extent possible, although this proved challenging in some cases, as detailed in the respective field reports, as well as adequate upstream and downstream

representation. Our engagement process ensured that participants represent and reflect diverse wetland user perspectives across both pilot sites. The reconnaissance visit was planned in close consultation with NBD and in-country National Discourse Forums (NDFs). Specifically, the following activities were undertaken:

- **Hotspot mapping:** Identify areas of visible stress (e.g., heavy siltation in the Mara, image courtesy: Radbound repository)



- **Access audits:** Ensure proposed data collection points are safely accessible by community representatives during both dry and rainy seasons.
- **Stakeholder identification:** Identify "hidden influencers" local elders or traditional leaders who may not be in official government roles but whose "blessing" is needed for community participation.
- **Capacity and technology assessment:** Assess existing capacity gaps, technology preferences (e.g., mobile tools, WhatsApp usage), and opportunities for strengthening citizen-led data collection. This included understanding community readiness, digital literacy levels, and availability of infrastructure such as mobile networks and power sources.

During the reconnaissance, a variety of stakeholders were met and engaged. Details of which are available in the following subsection.

2.4 Stakeholder identification

Informal meetings with local communities on site and structured engagement with the stakeholders (referred to as community and technical debriefing sessions) were combined during our reconnaissance visit. We mapped the key stakeholders in consultation with NBD and updated the list of stakeholders during our visit. We engaged three types of stakeholders during our visit

- **Strategic (National/Basin):** NBD Secretariat, country NDFs and Lake Victoria Basin regional offices/water boards to ensure that data feeds into policy.
- **Operational (Local government):** Relevant County (Kenya) and District (Tanzania/Uganda) officials to align our assignment with appropriate management actions/local environmental enforcement laws.
- **Grassroots (Local community, CBOs and NGOs):** Individuals, community based organisations (e.g. water user associations) and NGOs (both local and INGOs active around these sites). The idea was to ensure that the citizens not only generate data, they own it too.

2.5. Stakeholder engagement cum debriefing

Following the reconnaissance visits to the potential sampling sites, Akvo and NBD co-convened structured debriefing sessions with the local community and technical personnel. During the **community debrief sessions**, engagement was primarily conducted with the grassroots stakeholders. Preliminary field observations were shared from the field, validated observations with local people, and gathered additional insights/feedback from community members.

During the **technical debriefing sessions**, we involved strategic and operational stakeholders. In a few cases, community members were invited and grassroots NGO representatives to these sessions to ensure continuity of conversations and build trust among the key stakeholders. Mainly county/district government officials, relevant line departments (e.g. agriculture, meteorology), National Discourse Forums (NDFs), and other institutional stakeholders were present during the technical debriefing. The sessions focused on reviewing findings from the field, aligning with policy and regulatory frameworks, seeking feedback on our methodology and identifying implications for day-to-day operations and strategic decision-making. Details about these debriefing sessions will be available in respective field reports.

Stakeholder engagement and debriefing will continue beyond these preliminary meetings with the key stakeholders. Continued briefing will be undertaken to the local community and government officials through online (see section 2.10.1 on data workflows for details) and offline modes of communication.

Representatives from Akvo's partners (KEWASNET and UWASNET) will attend monthly community meetings (*Barazas*) and relevant local government officials to share curated monthly data summaries and seek their feedback. Once adequate data is available for consolidation and pattern recognition, learning sessions will be convened within the country/cross-country (exact nature of these sessions and the stakeholders to be engaged are yet to be discussed) to influence policy and practice.

2.6 Selection of sampling sites

Four sampling sites were selected per wetland- one upstream of the wetland (within the main river and/or a key tributary draining into the wetland), two locations within the wetland and one downstream of the wetland before the river drains into Lake Victoria. This approach ensures consistent comparison of upstream inputs, in-wetland processes, and downstream conditions across both pilot locations.

Sampling sites were selected based on:

- **Scientific criteria:** Points upstream (control), mid-stream (siltation/farming impacts), and downstream (accumulation of pollutants).
- **Social criteria:** Proximity to community water points, livestock watering holes, and papyrus harvesting zones.
- **Accessibility:** Within a 30 minute walk/ride for community representatives to ensure proper reporting.
- **Safety and security considerations:** To minimize risks to community monitors, including avoiding areas with high wildlife presence (e.g., crocodiles and hippos), unsafe terrain, or

locations associated with socially sensitive or restricted activities. Community acceptance and safe access will be prioritized.

- **Gender and inclusion considerations:** Accessibility and safety for women and other vulnerable groups, ensuring that monitoring locations are inclusive and do not disproportionately limit participation. Consideration was given to sites relevant to resources primarily used or managed by women.

Through these sampling sites in each wetlands, a **hydrological transect** shall be established across the Mara and Sio-Siteko Wetlands. This will allow us to compare "water in" vs. "water out" and whether the wetland is playing a role in improving the quality of water (physicometrical, ecological and hydrological). That brings us to the selection of quality parameters.

2.7 Selection of wetland health parameters

To provide a holistic view of wetland health, monitoring will be undertaken across the three data types on a monthly basis over six months. In addition, we will create a mechanism for common citizens to report on pollution incidents as and when they witness an event (e.g. industrial/sewage discharge, storm event, overgrazing etc.). We will use water quality tests to capture citizen attention. But our monitoring will not be restricted to water quality measurements alone. It will include measurements of ecological (e.g. Papyrus density), hydrological (e.g. water levels) and livelihood (e.g. fish Catch Per Unit Effort) indicators. In keeping with the basic premise of simplicity for citizen-led monitoring and the nature of ecology-linked livelihoods in the region, we have decided to place the livelihood indicator within the ecological category. The table in the following page describes the details.

Category	Specific parameters	Tools
Physico-chemical	pH, Temperature, Dissolved Oxygen	Handheld multi-parameter probes
Ecological	Papyrus density, Bird sightings (Cranes), Invasive floral species, fish Catch Per Unit Effort (CPUE)	Visual observations, photos, recording feedback from actual user communities
Catchment/hydrological	Any unusual human activity- cattle overgrazing, encroachment, mining, storm events, sewage/industrial discharge, choking of wetland inlets, water levels/flow	Visual observation and recording, Mobile app / GPS

In addition to these parameters, our academic partners will test for heavy metals and nutrient loads (N/P) in the water samples during the process of laboratory validation (see section 2.9 for details).

2.8 Selection of community representatives

Key stakeholders will be mapped as per their influence and interest in the wetlands. We will select the citizen scientists from the existing water user associations across the wetland sites and ensure that these citizen scientists represent most quadrants of the stakeholder map. A draft job description for citizen scientists is available in **Annexure 3**. In addition to the citizen scientists, we

will recruit enthusiastic young women and men from the local communities around these wetlands as “wetland watchers”. These women and men will be recruited via word-of-mouth publicity during monthly *Barazas* and targeted awareness campaigns using mobile service providers around the wetlands. A set of gender-based thumb rules will be applied to guide our hiring of enumerators and wetland watchers (see **Annexure 4** “gender and inclusion strategy” for details). They will be expected to act as an extended set of eyes and ears of the citizen scientists and report any unusual event as and when necessary (e.g. industrial/sewage discharge, storm event, overgrazing etc.). A draft reporting form for wetland watchers is available in **Annexure 1**.

2.9 Capacity building of citizen scientists and wetland watchers

Before commencement of field data collection, selected citizen scientists and relevant NBD/NDF focal persons will undergo structured training. We will also orient local communities to report pollution episodes and storm events. The training will be delivered in two phases:

- **Training of Trainers (ToT):** selected personnel from NBD, NDFs, KEWASNET, UWASNET will be trained in a workshop setting
- **Field training for citizen scientists:** site-based practical sessions on water quality monitoring, capturing data on biodiversity and hydrology, QA/QC procedures, and sampling protocols. Citizen scientists will be using smartphones to capture the data.
- **Orientation of wetland watchers:** local communities from the upstream country (Kenya) will be trained to report pollution episodes and storm events in the Mara and Sio rivers. Wetland watchers will be able to use any phone including feature phones to report pollution episodes.

The training will be conducted **sequentially**, with ToT preceding field level training. The orientation will be carried out in **parallel** to the training.

2.10 Validation of results

The results will be triangulated through a participatory validation process, whereby local communities around the wetlands will peer review data generated by their fellow citizens during the monthly meetings. In addition, our academic partners will also collect samples from the selected sites, analyse those samples in the laboratory and generate their own results in order to verify the numbers generated by the citizen scientists during on-site testing.

Curated data summaries will be developed from the monthly water quality testing exercises and citizen reports from the wetland watchers. These summaries will be available online for public consumption. In addition, data summaries will be actively shared with the local communities during monthly *Barazas* and relevant local government officials during one-on-one interactions. During these meetings/interactions, we will seek feedback/opinions from the key stakeholders on the water quality test results. Opinions won't change data but they might shine light on a few important insights that the data may or may not reveal. This will be one type of data validation.

Every 3 months, our academic partners will collect water samples from the selected sites to test for all the parameters that the citizen scientists have been testing for. In addition, they will test for **Biochemical Oxygen Demand, Orthophosphate, Nitrate, and Mercury** in the water samples. Our academic partners will develop a validation protocol and share the results of validation with the citizen scientists to improve the field-testing process. Quality assurance/control protocols for

validation are described in detail in **Annexure 5**.

In addition to monthly water quality datasets, the validation process will incorporate citizen reports, indigenous knowledge, community observations from *Barazas*, and incident reports from the Kenyan side of Mara and Sio rivers. This ensures that validation is not limited to laboratory and summary statistics alone, but also reflects lived environmental realities and community feedback loops.

2.11 Data workflows and technology architecture

Akvo will build the online data platform following open source licenses(for the custom components) and open knowledge systems. We will build a foundational platform for this assignment, which can be further developed if NBD decides to add new data layers during the scale up phase.

2.11.1 Data workflows

The data platform will be structured around three layers: data collection (input), administration, and visualisation (output). The platform will support two distinct workflows that reflect the different roles, geographies, and data types involved in this assignment.

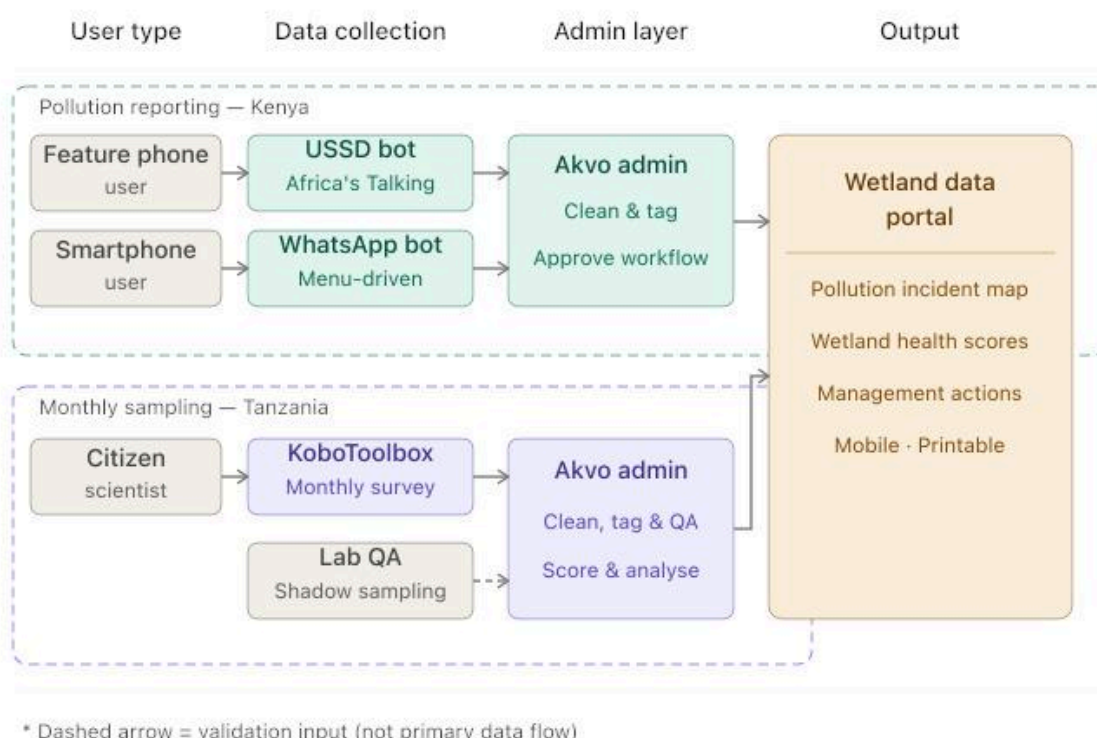
The first workflow covers pollution episode reporting and operates in Kenya. Citizens report unusual events, industrial discharge, sudden changes in water colour or smell, flooding, encroachment, as and when they occur. Feature phone users access a USSD-based reporting menu, while smartphone users submit reports through a WhatsApp Business bot with a structured, menu-driven flow.

Both channels are designed to minimise barriers: no smartphones, no apps, and no internet connectivity are required for USSD submissions. The data collected through this workflow feeds directly into the admin layer, where Akvo staff clean, tag, and approve each submission before it is published to the output portal. The estimated maintenance cost for the platform (integrations, data portal etc) including USSD and Whatsapp aggregator costs are expected to be about 500 USD per month. We are currently in discussion with the appropriate service providers(s) to finalise the arrangement. Over time, the costs will be further optimised and brought to a minimum.

The second workflow covers structured monthly water quality sampling and operates in Tanzania. Trained citizen scientists visit designated sampling sites once a month, collect physico-chemical measurements, ecological and hydrological observations, and submit structured data via KoboToolbox. Submissions are reviewed and scored by Akvo staff, and results are periodically validated through shadow sampling/ formal laboratory validation conducted by our academic partners. The scored outputs are then integrated into the output portal alongside the pollution incident data.

Where available and appropriate, the platform may also draw on relevant secondary datasets, such as existing wetland baseline records and publicly available Earth Observation data, to supplement citizen-generated observations. This is not a primary data integration pipeline but a means of adding context where citizen data alone is insufficient to compute a parameter score.

For Earth Observation data, we intend to use raw images from Sentinel 1 and 2 along with CHIRPS. We will set up a data processing pipeline (automated jobs set up), which will help to augment as well as help in validating the ground observations.



The citizen-generated data, supplemented (where needed) by secondary sources, is used to compute wetland health scores at the parameter level, aggregate them into a single composite score, and assign a health class to each wetland (see **Annexure 2** for details). Indigenous knowledge captured through community interviews adds qualitative context to these scores and informs how borderline results are interpreted.

Once scored and validated, the data informs a set of management actions structured around a traffic light classification. A green status indicates no immediate action is required. A yellow status triggers a local community response. A red status results in a formal report to NBD and the relevant local or national government authority for enforcement action.

The following table summarises the key steps in the data workflow.

Steps	Details	Responsible actor(s)
Capture	USSD and WhatsApp bot for pollution reporting; KoboToolbox for monthly sampling	Akvo, KEWASNET, UWASNET, academic partners
Validate	Quality checks through field verification, shadow sampling, and CSO oversight	Akvo

Integrate	Coordination of monthly sampling, shadow sampling, and CSO oversight	KEWASNET & UWASNET
Visualise & analyse	Wetland health scores, geospatial incident maps, and management action summaries	Akvo
Publish	Outputs made available via the wetland data portal in mobile and printable formats	Akvo

2.11.2 Technical architecture

Platform vision

A mobile-first, citizen-centric data platform enabling community-driven data collection, administration, and visualisation through familiar and accessible tools, with two purpose-built workflows that converge into a single public-facing output portal.

Open-source foundation

The architecture prioritises open-source components wherever possible to ensure transparency, reduce licensing costs, and support long-term sustainability and institutional ownership. Proprietary components are limited to the two citizen-facing communication channels, WhatsApp Business API and USSD via Africa's Talking, both selected because of their dominant adoption in the target communities and their ability to reach citizens without requiring smartphones, internet data plans, or app installations.

Data collection layer

WhatsApp Business API (proprietary) serves the pollution reporting workflow for smartphone users. Citizens interact with a structured, menu-driven bot that guides them through the submission of a pollution incident, selecting the incident type, sub-county, and optionally attaching a GPS-stamped photo or voice note. The menu-driven design reduces input errors and eliminates the need for free-text interpretation by staff.

USSD via Africa's Talking (proprietary) extends the pollution reporting workflow to feature phone users. Citizens dial a dedicated short code and navigate a branching menu to report an incident by type and location. Every session closes with a message confirming receipt and providing the contact details of the local KEWASNET representative. Because USSD runs over the GSM voice network, it functions in areas with no mobile data coverage, ensuring that the most remote and resource-constrained community members can participate.

KoboToolbox (open source) supports the structured monthly sampling workflow. Citizen scientists use the mobile app to complete a geo-referenced, offline-capable survey at each designated sampling site. The form covers physico-chemical measurements, ecological observations, and photographic documentation. KoboToolbox was selected for its proven track record in humanitarian and development contexts, its robust offline functionality, and its native support for XLSForm standards.

Admin layer

The two workflows maintain separate processing pipelines in the admin layer before converging at the output stage. The pollution pipeline focuses on cleaning, tagging, and approving incoming incident reports through a structured review workflow. The sampling pipeline involves a more intensive process of data cleaning, QA review, parameter-level scoring, and aggregation into composite wetland health scores.

Interoperability

The platform is designed for interoperability with national and international data ecosystems. Sampling sites are assigned persistent, structured identifiers at the reconnaissance stage — for example, **NBD-MARA-001** — which are referenced consistently across all data layers and external data exchanges. Spatial layers are published as GeoJSON and accessible via documented API endpoints, enabling NBD, county governments, and other authorised stakeholders to query the data programmatically. Metadata follows ISO 19115 profiles to support harvesting by national and regional open data catalogues. These alignments also ensure that aggregate wetland health scores can be mapped to SDG 6.6.1 sub-indicators without manual rework.

Output portal

The two pipelines converge in a single wetland data portal. The portal presents pollution incidents on an interactive geospatial map, wetland health scores at parameter and aggregate level, and management actions linked to the traffic light classification. All views are accessible on mobile devices and can be exported in a printable format (PDFs) for offline dissemination at *Barazas* and government meetings.

Deployment

The platform is fully Dockerised, ready for local in-country server hosting, and requires minimal hardware specifications. This ensures that NBD and its national partners can operate and maintain the platform independently after the pilot period, without dependence on external cloud infrastructure.

2.12 Activities completed so far and remaining activities

Activities	Responsibilities	Completed	Timelines
Stakeholder identification and consultations	NBD	✓	Jan 2026
Reconnaissance field visits and debrief sessions with the community and technical personnel	NBD & Akvo	✓	Feb- April 2026
Developing a comprehensive technical and conceptual framework and platform including the data collection methods and tools, data storage, system development	Akvo	x	May-June 2026

and maintenance,data analytics, governance issues			
Development and deployment of scalable system, testing its functionality and initial hosting of the system during the project period	Akvo	x	May-June 2026
Select stakeholders (including NBD and relevant NDFS) and strengthen their capabilities on the methods and tools for data collection, analysis, processing, management, and usage.	Akvo & NBD	x	May – June 2026
Document the data collection, processing, managing and utilization processes; identify challenges, successes, and prospects; develop a clear guiding manual/toolkit for scaling-up the model across different catchments in the Nile basin; and handover the process to NBD for replication and scaleup.	Akvo	x	July - December 2026
Scale-up phase:includes providing coaching and mentoring support during initials scale- up	Akvo & NBD	x	October 2026 – Jan 2027
Project completion		x	Jan – March 2027

3.0 Detailed timeline (15 Months)

This timeline presents the project's 15-month phased implementation across key workstreams: inception, co-design and development, implementation and validation, and replication and scale-up, with project management running throughout. It includes detailed activities from initial research and stakeholder engagement to citizen-led data collection, monitoring, community validation, and knowledge sharing. We have used an image of the timeline in order to fit it into a single page. Please feel free to Annexure 6 for a closer look at the detailed activities and the corresponding timeline.

Workstream	Activity	2026												2027		
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
01 - Inception	Desk research															
	Inception discussions and report preparation															
	Inception report finalisation															
	Mara wetlands reconnaissance visit to map hotspots, identify stakeholders and start initial engagement with stakeholders															
	Draft site visit report - Mara															
	Finalisation of Mara site visit report															
	Bio Sileko reconnaissance visit to map hotspots, identify stakeholders and start initial engagement with stakeholders															
	Draft site visit report - Bio Sileko															
	Finalisation of Bio Sileko site visit report															
	Make a shortlist of potential sampling sites for finalisation															
	Draft technical Design and data architecture- clarifying data flow, validation checkpoints, and data sharing															
	Finalisation of technical Design and data architecture- clarifying data flow, validation checkpoints, and data sharing															
02 - Co-design and framework development	Set up formal partnerships with sub-consultants within the project consortium (including final decision on the academic partner for the Mara wetlands)															
	Set up formal and informal partnerships with relevant NGOs/ government institutions outside the consortium															
	Develop a draft system design document of the data platform															
	Finalise the system design document in consultation with NBD															
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	Develop a draft sampling and testing protocol															
	Finalise the sampling and testing protocol															
	Develop a draft QA/QC protocol for validating the on-field test results and reports															
	Finalise the QA/QC protocol for validating the on-field test results and reports															
	Develop a draft of the wetlands health report card format															
	Finalise the report card in consultation with NBD															
	Build a prototype of the data platform															
03 - Implementation and validation	Design and deploy the MVP of the data platform															
	Finalise sampling sites in consultation with NBD, KEWABNET, UWABNET and academic partners															
	Select citizen scientists in collaboration with NBD, relevant NDFs, KEWABNET, UWABNET and community representatives															
	Conduct Train-the-Trainers (ToT) workshops for UWABNET, KEWABNET and NBD/NDF staff in a workshop setting															
	Use the trainers to train citizen scientists on-site (1 training per wetland) to collect water quality, ecology, hydrology and observational data															
	Prepare draft training reports															
	Finalise training reports															
	Develop draft sampling and testing protocols and guides for observations															
	Finalise the sampling and testing protocol and guides for observations															
	Data (water quality, ecology, hydrology and observational) collection by the citizen scientists															
	Package awareness campaign messages in collaboration with KEWABNET															
	Design workflows within the data platform to collect citizen reports and use reporting results to spread awareness															
	Orient "wetland watchers" to report pollution incidents and storm events															
	Run awareness campaigns for upstream catchment communities in Kenya															
	Incident reporting and sharing of key findings from wetland watchers (offline channels- word-of-mouth, monthly Barazas, one-on-one meetings)															
	Ongoing monitoring, oversight, and technical support during data (water quality, ecology, hydrology) collection, sharing and pollution reporting															
	Data validation by citizens															
	Data validation by academic partners															
04- Replication and scale up	Data management- curation, analysis and visualisation for wetland health report cards															
	Development of a replication toolkit, lessons learnt document															
	Design support during scale-up planning															
05- Project management	Facilitate learning and dissemination events in consultation with NBD, relevant NDFs, key government, non-government and donor partners (sharing lessons within and between countries)															
	Project management															

4.0 Risks to delivery and timelines

The following matrix identifies the specific hurdles for the Mara and Sio-Siteko basins and provides pre-emptive suggestions for mitigating the risks.

Risk category	Potential threats	Mitigation suggestions	Risk owner(s)
Political	Sensitivity regarding transboundary data (e.g., pollution sources).	Use the data sharing agreement framework; focus on "Wetland health" rather than "Blame."	NBD and Akvo
Technical	Phone loss, damage, or lack of charging in remote areas.	Paper logbook Backups; provision of solar power banks to community scouts.	Akvo/ KEWASNET/UWASNET
Social	"Elite capture" (data only reflects the views of powerful leaders or tech savvy population).	Monthly <i>Barazas</i> where data is read aloud for public validation.	KEWASNET and UWASNET
Data Integrity	Citizen scientists fudging data to finish quickly.	GPS-stamped photos; 10% random shadow sampling by the academic partners.	Academic partners
Environmental	Seasonal floods making sampling sites inaccessible.	Flexible timelines; identification of safe sampling points during reconnaissance.	Akvo
Economical	Loss of interest of data collectors	Appropriate incentive mechanisms and motivation, careful selection approach	Akvo and NBD

5.0 Success indicators

Success of our interventions will be defined following a results matrix. Another simple metric we intend to use is whether our work has lived on beyond the duration of this assignment. An additional success metric will be the quality of data used to inform strategic and operational decision-making processes.

5.1 Results matrix

At a high level, we will measure our performance through the following outputs and outcomes:

Outputs	Outcomes
• Number and type of stakeholders identified • Number of trainings, refreshers and mentoring sessions held for citizen scientists • Type of survey instruments (questionnaire and guides) and support materials created • QA/QC protocol developed for data validation • Number of data dashboards/summaries created and insights generated • Whether the technology deployed to facilitate the data workflows is easy to replicate and relevant to the context	• % of wetland management plans that are using citizen generated data • % of wetlands utilizing insights from data to draft/revise policies, allocate resources and make conservation decisions • % of wetlands in the Nile basin that are interested to use the technologies, systems, processes developed in this pilot to build their own data ecosystems

5.2 Sustainability

While sustainability is a journey and not a destination, measures will be put in place to ensure that we have designed, measured and learned from our progress in this journey. A set of processes will be embedded in our day-to-day programming to ensure that our work lives on after the 15-month contract ends:

- **Adaptive design, no vendor lock-ins:** Rather than prescribing a pre-selected software suite, Akvo will identify and configure the most appropriate open-source components through user testing and feedback loops. This ensures usability, affordability, and institutional ownership. We will document, containerise, and fully transfer the platform to NBD and its partners after the pilot, avoiding vendor lock-in and securing long-term sustainability.
- **Institutionalization:** We will lobby for the "citizen scientist" role to be formally recognized in the annual budgets of the **County (Kenya)** and **District (Tanzania, Uganda)** water resource/environmental offices and the contributions of these citizen scientists are celebrated on Mara day (15th September) every year for Kenya and Tanzania.
- **Knowledge transfer:** A "Training of Trainers" (ToT) manual will be produced in local languages (Samia/Swahili) to allow new citizen scientists to be trained without external consultants.
- **Toolkits for replication and support for scale up:** Akvo will develop easy-to-use toolkits for replicating this assignment beyond the scope of this pilot project. In addition, we will also build capacity for NBD staff and support the design for scale up in the relevant river basins.

Annex 1: Citizen reporting form

This annex presents the citizen reporting form designed for citizen scientists and wetland watchers to capture key environmental observations in a simple, standardized format. The tool uses a traffic-light system to record changes in water quality, river flow, vegetation, biodiversity, and human activities, supporting consistent and accessible data collection across monitoring sites.

Reporter/wetland watcher name (optional): _____

Location/Site: _____

Date: / /2026

Water clarity: Clear (Green), Murky (Yellow), Dark/Smelly (Red)

River Flow: Normal (Green), Slow (Yellow), Stopped (Red)

Water levels: High (if the river banks are not visible and floodplains are inundated), Medium (if the banks are visible and floodplains are not inundated), Low (if the water has trickled down a small stream/narrow river and parts of the river bed is visible)

Nature & Plants (Check one) - Papyrus: Thick (Green), Burnt/Cleared (Yellow), Gone/Muddy (Red)

Cranes/Birds: Seen Daily (Green), Rarely Seen (Yellow), Not Seen (Red)

Human Activities (Check one) - Farming: Far Away (Green), Near Edge of wetlands (Yellow), Within the waterbody (Red)

Mining: Small pits (Green), Many pits (Yellow), Banks falling (Red)

Voice notes/Text comments (Optional):

Annex 2: Wetland health report card template

1. Introduction

This annex presents a wetland health report card template designed to compile and synthesize environmental, ecological, and community-based observations into a structured assessment of wetland condition. The tool integrates quantitative indicators with qualitative insights, including indigenous knowledge on changes in flooding patterns, fish spawning, and vegetation cycles, to provide a holistic understanding of wetland health. Based on its condition and aggregate scores, wetland health can be classified under the following categories:

- A - Very Good/Natural: Unmodified or natural, with very high ecological integrity. Score 0.8-1.0
- B - Good/Slightly modified: Largely natural with few modifications; small loss of natural habitat. Score 0.6-0.8
- C - Moderate/Moderately modified: Moderate change in ecosystem processes and loss of natural habitats. Score 0.4-0.6
- D - Poor/Largely modified: Large change in ecosystem processes, with serious loss of natural habitat and biota. Score 0.2-0.4
- E - Very poor/Critically modified: Critical modifications, with ecosystem processes completely altered (e.g., severe draining or pollution). Score 0.0-0.2

2. General information

Name: Insert name		Map:	Photo (use time stamped camera while taking photos)
Area:	Land use:	Dependent population:	Different uses of the wetland:
What are the different types indigenous knowledge regarding the wetland:			
Have the floods in and around the wetland become more regular over the last 5 years (captured through anecdotes like water levels, cloud movements, bird and insect behaviour, unusual flowering or			

shedding of leaves, wind speed etc.)?)	
Have the fish-spawning periods in the wetland changed over the last 5 years (captured through anecdotes like water level, physical appearance of water, fish migration patterns etc.)? If yes, how?	
Have the Papyrus growth cycles of the wetland changed over the last 5 years (captured through anecdotes like harvesting patterns, growth rates and seasonal fluctuations)? If yes, how?	
What are the major threats (natural and anthropogenic) to the wetland according to the local community?	

WETLAND HEALTH REPORT CARD



Location Wetland area

Name of the wetland

General information (description)

Land Use

data

Wetland Use

data

Dependent population and Households

data

Major threats (Natural and Anthropogenic stressors)

data

Management regime

data

3. Wetland status

We shall use a structured scoring framework that aggregates key indicators across physico-chemical, ecological, hydrological, and governance dimensions to determine the overall condition of the wetland.

Parameters	Score*	Group Mean	Aggregate score	Wetland Class
Physico-chemical: Water temperature, pH, Dissolved Oxygen Converted into a Water Quality Index score	0.27	0.27	0.34	D
Catchment and hydrological				
% wetland converted to non-wetland use	0.8	0.65		
Ratio of number of natural inlets choked and diverted to total number of natural inlets	0.5			
Ecological				
% wetland area covered by invasive macrophytes	0.25	0.45		
Catch per unit effort	0.6			
Annual waterbird count as a proportion of average count of last 5 years	0.5			
Governance: Whether a wetland management plan was available and approved by the appropriate authority	0	0		

***Note:** Dummy numbers to show how the scoring process would work. Final score will be adjusted⁴ based on contrasting the numbers with qualitative information (indigenous knowledge) about the wetland captured through community interviews. An explanatory note on fuzzification/fuzzifying indigenous knowledge is available at the end of this annexure.

⁴The purpose of the indigenous knowledge captured through the community interviews is generally to add a meaningful context to the data and make it easier to interpret the findings. But over and above the usual practice, we intend to use some form of fuzzy logic to embed experiences into the scoring process. We will convert the group scores into fuzzy sets, establish appropriate “if-then” rules, process inputs and apply fuzzy rules to the data. We will aggregate results of all the relevant rules into a single fuzzy output set and use a defuzzification algorithm to convert the fuzzy output into a precise numerical value- the adjusted aggregate score for the wetland

4. Management actions identified⁵

The wetland X is at a poor status/class (D). Water quality parameters contribute heavily to this low score. This low score can be attributed to unregulated industrial effluent and sewage discharge, which has led to a poor dissolved oxygen and high biochemical oxygen demand. In addition, a significant portion of the wetlands has been converted to non-wetland uses. Which has also affected the biodiversity and catch per unit effort. There isn't a proper wetland management plan available either. It needs the following actions:

- Prompt reporting effluent and/or sewage discharge incidences at the local unit of Ministry of Environment and Water. Phone number XXXXXX
- Setting up interceptor STPs to direct sewage to sewage treatment plant/combined effluent treatment plant. A plan including the budget and timeline estimates will be presented in the meeting dated xx.xx. 2026
- Preventing any further encroachment of the wetland via enacting protected area/buffer zone regulation (e.g. increasing buffer widths) as per the Environment Management & Coordination Act. Any violation should be reported immediately to the County Environment Committee. Phone number XXXXXX
- Fishermen should switch to better, lower meshed gillnets to increase the catch per unit effort
- The Environment Management Authority should put together a draft management for Wetland X for public review by December 2026

5. Specific stakeholder level actions

Water user associations: Keep an eye on the growth of local invasive species, do not discharge sewage or throw trash directly into the wetlands, report pollution incidents immediately

Country/District government or below: Record livelihood impacts more frequently (e.g., drop in fish catch), allocate resources (e.g., providing seedlings for buffer zones, subsidising new fishnets, funding a water hyacinth cleanup campaign) and enforce by-laws according as per the mandates of the County Environment Committee.

National government: Identify transboundary impact, ensure alignment with national targets (e.g. SDGs), review policies and infrastructure investments (e.g., interceptor STPs), ensure coordination across counties/districts.

6. An explanatory note on fuzzification

I. Harmonized linguistic Scales

To prevent subjective interpretation, we will map community qualitative inputs (e.g., observations on biodiversity, water flow, and vegetation) onto a standardized linguistic scale. Each term in this scale will be assigned a fixed Membership Function within a defined Universe of Discourse. This

⁵A set of imaginary yet indicative actions described based on the class of the wetland and management actions. In reality, the observations, parameter scores and wetland class should be discussed together in a series of meetings before a more nuanced set of management actions (as per the stakeholder group) are identified and agreed upon

ensures that a "moderate decline in water levels" reported in the Mara is mathematically equivalent to a "moderate decline in water levels" reported in Sio Siteko.

II. The universal fuzzy rule base

The "if-then" logic used to process inputs will be governed by a master logic matrix. This matrix is developed by cross-referencing indigenous ecological indicators with established scientific wetland health parameters. By keeping the rule base static across different datasets, we eliminate the risk of analyst bias and ensure that the scoring process remains consistent, regardless of the team performing the calculation.

III. Inter-rater reliability & calibration

Before the final scoring, we will conduct a calibration exercise using a subset of the interview data. We will apply a centroid defuzzification method to convert fuzzy outputs into precise numerical values.

By anchoring our fuzzy logic in these pre-defined mathematical structures and fixed rule sets, we shall ensure that the indigenous knowledge is not just added context, but a standardized, verifiable component of the wetland's health score.

Annex 3: Selection and job description for a citizen scientist

Role: Citizen scientist
Location: Sio Siteko and Mara Wetland Catchment areas
Reports to: Community engagement specialist of UWASNET/KEWASNET
Commitment: Part-time, Ongoing (Estimated 4-8 hours per month)

1. Selection process

The ideal candidate is within the age group of 20-50 years, representing water user associations, self-help groups or farmer cooperatives or an interested individual who is not part of any collective. Candidates representing disadvantaged groups or women will be especially encouraged to apply. In case there is an existing citizen science programme in the region (e.g. Mara wetlands), we will try to reuse these citizen scientists in our assignment too to avoid duplication of efforts and ensure efficiency. In other cases, our NGO partners KEWASNET and UWASNET will meet the tribal chiefs around the selected sampling sites and choose the citizen scientists in consultation with the chief/village head.

2. Role summary

The citizen scientist is a dedicated community member responsible for the routine collection of essential ecological and social data within their designated wetland area. This data is the foundation of the Wetland Health Report Card, directly informing the water user association, local and national government wetland management decisions and helping to secure the long-term sustainability of the Sio Siteko and Mara ecosystems.

3. Key responsibilities

The primary function is to serve as the project's "eyes and ears" on the ground. This involves:

A. Site selection support

Accompanying Akvo and other consortium members (university and NGO partners) during the reconnaissance, identifying anomalies across the wetlands and supporting selection of sampling sites.

B. Primary data collection and reporting

- Conducting basic water quality tests (e.g., pH, Electrical Conductivity, Dissolved oxygen) using provided handheld kits at designated sampling points.
- Collecting general data (e.g. photo, dependent population, identifying threats)
- Identifying and recording the presence and relative abundance of invasive species (e.g., water hyacinth) and reporting pollution incidents as and when an episode occurs
- Documenting and geo-locating signs of human disturbance (e.g., solid waste dumping, encroachment, illegal harvesting, pollution discharge points).
- Observing and recording water use activities by Water User Association members and other key stakeholders
- Capturing indigenous knowledge about the wetland through community interviews

S/he is expected to accurately observe, collect general data, measure water quality parameters, and take photographs using a standardized mobile data collection app.

4. Skills and qualifications

While formal education is not required, successful candidates will demonstrate:

- Residency: Must be a resident of the local community and have intimate knowledge of the designated wetland site(s)
- Reliability: Possesses a strong sense of responsibility, punctuality, and commitment to routine tasks.
- Literacy/Numeracy: Basic reading, writing, and arithmetic skills sufficient to accurately record data and follow technical instructions
- Commitment to Conservation: A genuine interest in and dedication to the health and sustainability of the local wetland environment and community livelihoods
- Communication: Ability to clearly communicate observations and potential concerns to the project team and community members.

5. Benefits and incentives

In return for their vital contributions, the Citizen Scientist will receive:

- Capacity building: Comprehensive training in basic wetland ecology, hydrology, water quality testing, monitoring protocols and mobile data collection tools. There might be opportunities to participate in livelihood improvement training too
- Stipend: A modest stipend to cover time and costs associated with data collection
- Equipment: Access to necessary field equipment (testing kits, safety gear, field notebooks, and potentially a mobile device/data bundle)
- Recognition: Official designation as a "citizen scientist" and inclusion in project meetings and community recognition events
- Direct impact: The data collected by the citizen scientist will be used to generate the wetland health report cards, directly influencing local government resource allocation and management policies

6. Who should and how to apply?

If you are passionate about making social change and are keen to contribute to environment/water resource management issues, we are looking for you!

Interested candidates should contact XX at UWASNET/KEWASNET directly over phone or WhatsApp at the following number XXXXXX.

Annex 4: Gender and inclusion strategy

To meet NBD and Akvo's focus on marginalized groups, we shall try to implement the "40-30-30" rule while selecting the citizen scientists or recruiting the wetland. We shall use these thumb rules to demonstrate our values. They are not an excuse for symbolism. We will not dilute our key criteria for citizen scientists and/or wetland watchers while aiming to implement the thumb rules.

- **Participation targets:** We will target 40% women, 30% youth (under 35), and 30% traditional elders as the core data generation team.
- **Context-sensitive scheduling:** The *Barazas* will be held between 10:00 AM and 2:00 PM to accommodate women's morning and evening household chores.
 - Child-Friendly spaces: Providing a safe space for children during Barazas to ensure mothers can participate fully.
- **Inclusive indicators:** Citizen scientists will be trained to collect data on resources primarily managed by women, such as domestic water quality and non-timber forest products (e.g., medicinal plants in papyrus zones).

The youth-elder bridge: Tech-savvy youth will operate the mobile devices, while elders provide the oral history/indigenous knowledge required to interpret "Yellow" or "Red" traffic light alerts.

Annex 5: Quality assurance & control (QA/QC) protocol

To ensure that "Citizen-Led" does not mean "Unreliable," we will use a three-step validation digital workflow:

Level 1 (The Device): Data collection forms will have **Logic skips** and **validation rules** (e.g., preventing a pH entry of 15).

Level 2 (The Community): The **Monthly Baraza** acts as a "human firewall." If the community disagrees with a data point, it is flagged in the system for a re-visit.

Level 3 (The Lab): Academic partners will conduct quarterly "shadow sampling," taking samples at the same sites as citizen scientists to compare results and calibrate the handheld probes

Annex 6: Detailed timeline

		2026												2027		
Workstream	Activity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
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	Finalisation of Mara site visit report															
	Sio Sitiko reconnaissance visit to map hotspots, identify stakeholders and start initial engagement with stakeholders															
	Draft site visit report - Sio Siteko															
	Finalisation of Sio Siteko site visit report															

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	Draft technical Design and data architecture-clarifying data flow, validation checkpoints, and data sharing																		
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	Orient "wetland watchers" to report pollution incidents and storm events																	
	Run awareness campaigns for upstream catchment communities in Kenya																	

[illegible]